Pituitary Abnormalities

Barbara Simon MD FACE
Barbara.Simon@Jefferson.edu

Clinical Associate Professor of Medicine
Division of Endocrinology

Learning Objectives

- Describe basic pituitary anatomy and physiology
- Identify and explain the clinical syndromes caused by pituitary adenomas and pituitary hormone hypersecretion and the basic approach to diagnosis
- Define and recognize the clinical syndrome of hypopituitarism
- Summarize the clinical syndromes of diabetes insipidus (AVP deficiency) and SIADH
- Review the basic approach to the pituitary incidentaloma

Overview

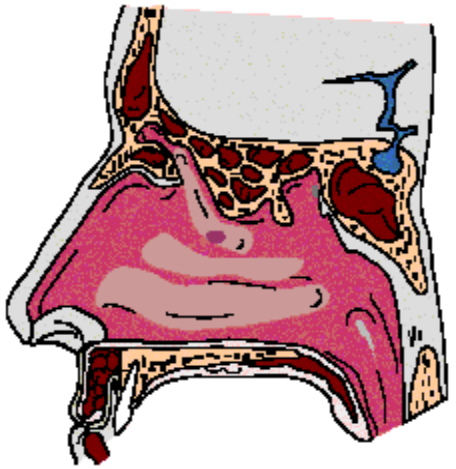
- Pituitary Anatomy
- Pituitary Imaging
- Sellar Enlargement

- Pituitary Adenomas
  - Individual hormones
  - Conditions of overproduction
- Posterior Pituitary

- Pituitary Underproduction/ Panhypopituitarism
- Incidentalomas

Pituitary- Anatomy

Located at the base of the skull- in the sella turcica (part of the sphenoid bone)

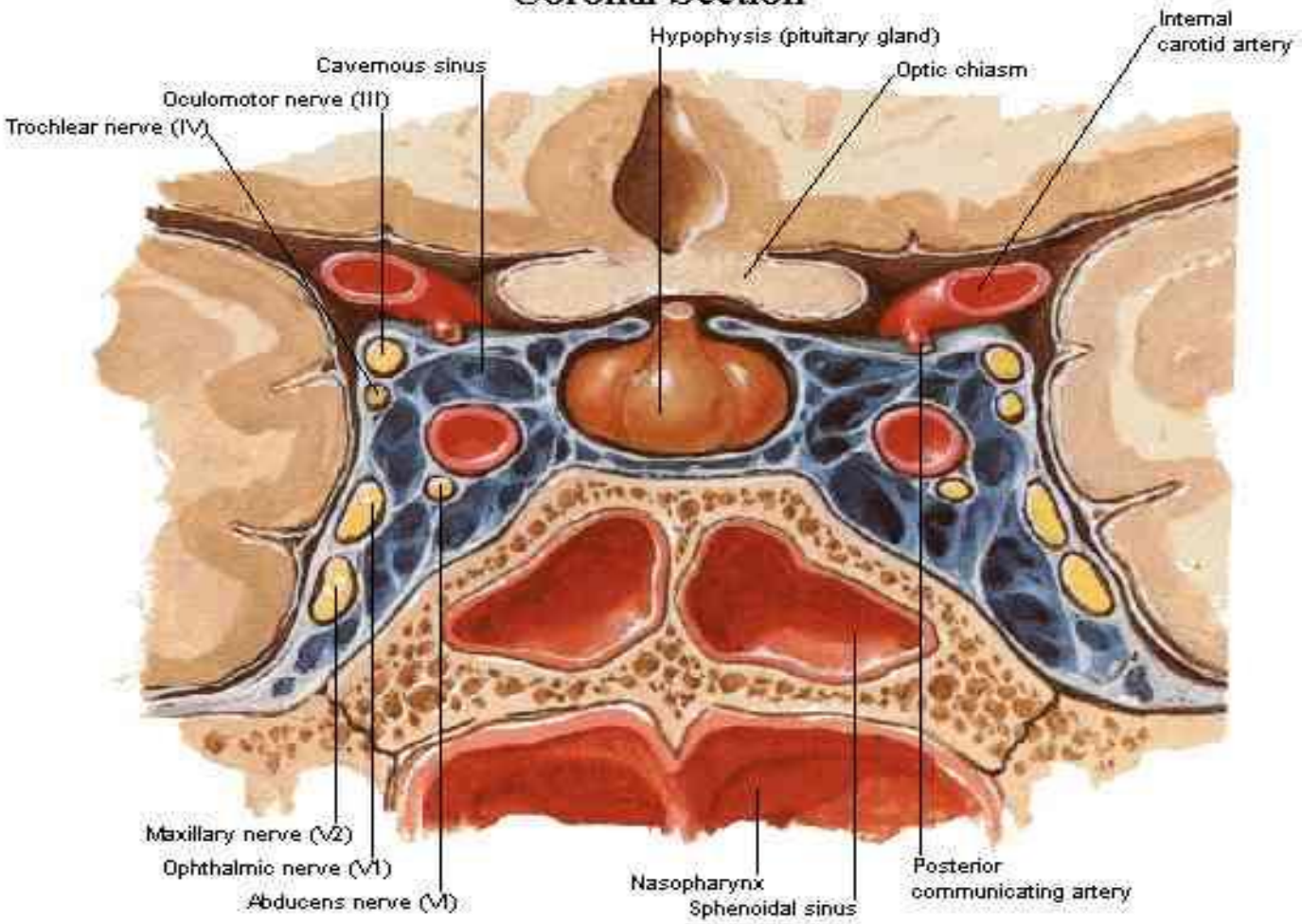


Critical structures nearby

- Optic chiasm (superior)
- Cavernous sinus (laterally)
  - Contains CN III, IV, VI
- Carotid arteries (laterally)

Cavernous Sinus

Coronal Section



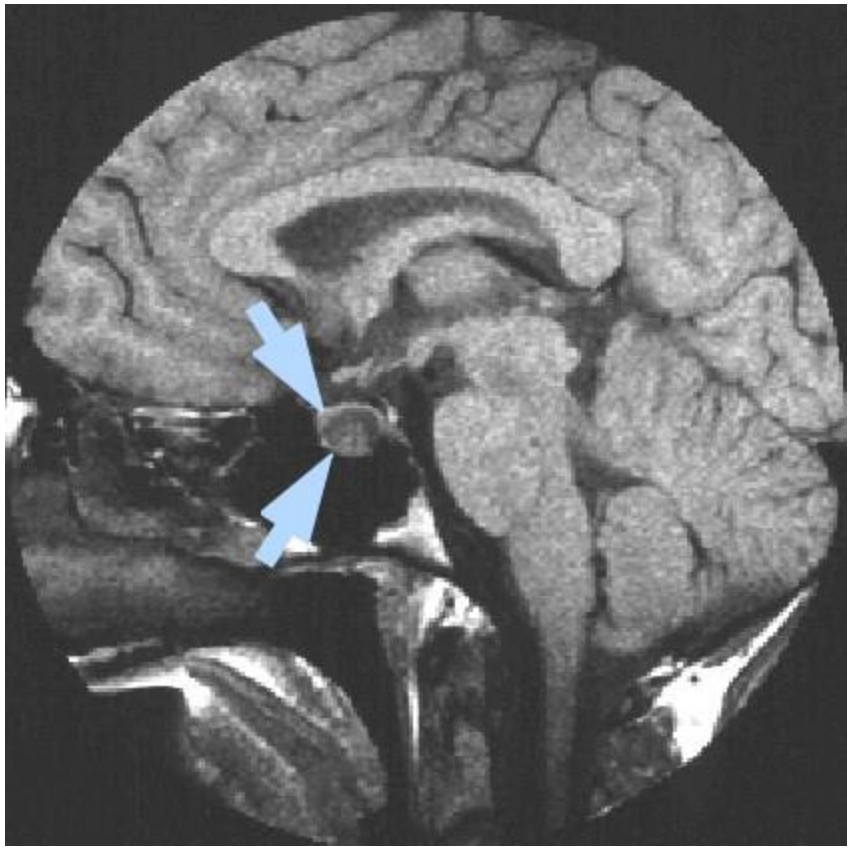
Pituitary Imaging

Basic principal of endocrinology-**confirm** that hormone secretion is abnormal before you look for the source….

GET THE BIOCHEMICAL DIAGNOSIS BEFORE YOU START IMAGING

10% of otherwise healthy people will have nonfunctional asymptomatic pituitary tumors (**incidentalomas**)

MRI is the most sensitive imaging modality



Sagittal View



Coronal View

Causes of Sellar Enlargement

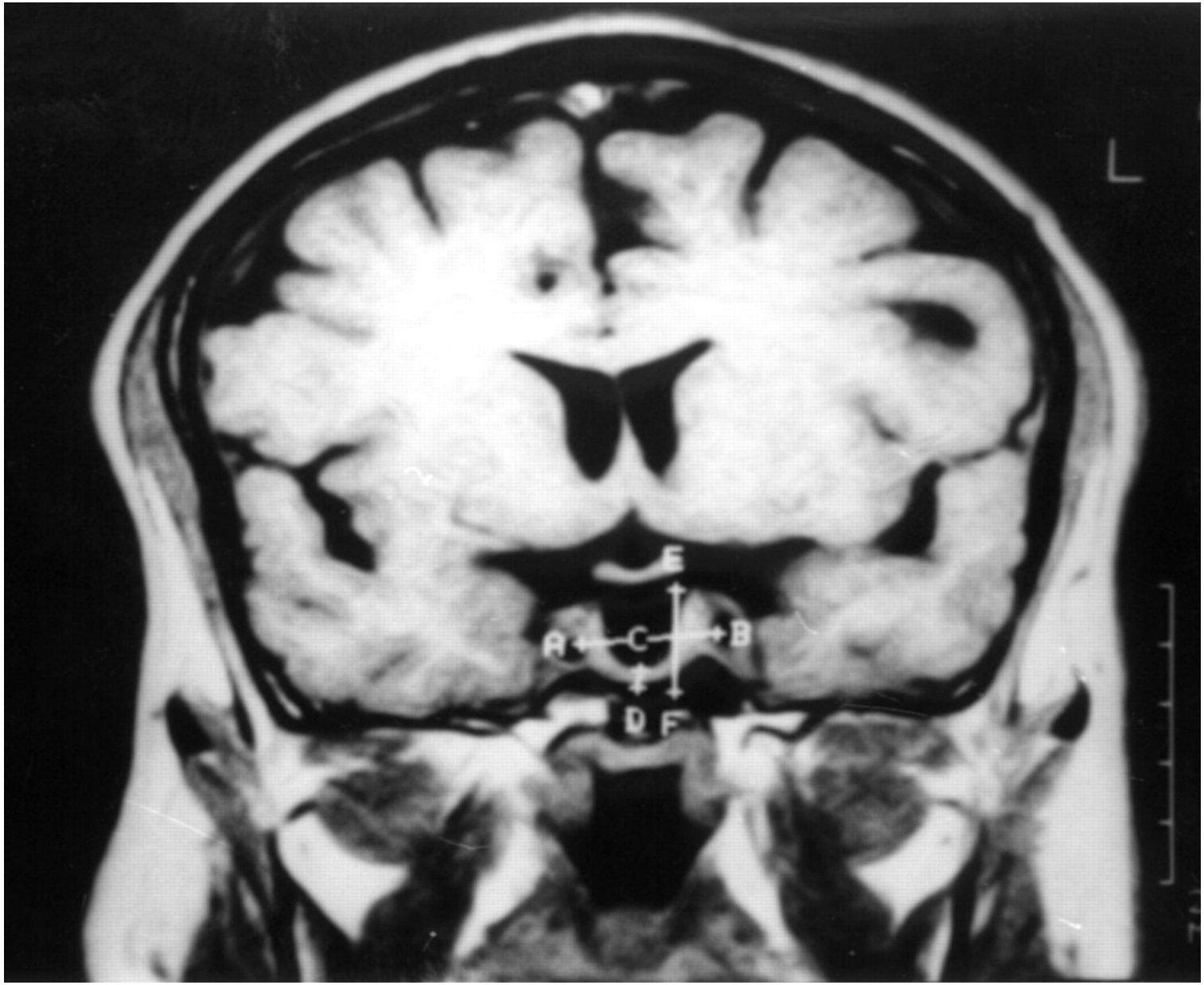
- Intrasellar tumor
  - Pituitary Adenomas
- Hypophysitis (inflammation)
- Parasellar Tumor (glioma, meningioma)
- Intrasellar aneurysm
- Hemorrhage
- Infiltrative disease (sarcoid)
- Empty Sella Syndrome

Empty Sella Syndrome

- Primary empty sella - defect in the diaphragm of the sella, thought to allow cerebrospinal fluid (CSF) to enlarge sella/ compress the pituitary
  - Usually no pituitary dysfunction
  - Sometimes elevates prolactin

- Secondary empty sella - association of empty sella with an identifiable disease of the pituitary gland
  - Example: pituitary adenoma that enlarges the sella but then is removed by surgery, radiation
  - May have pituitary dysfunction

- Evaluation of a patient whose MRI shows an empty or partially empty sella, depends on clinical situation….(ex. checking for hormone deficiencies in secondary, if any clinical symptoms)


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Pituitary Adenomas (& definition of terms)

Pituitary adenomas account for approx 90% of sellar tumors, virtually always benign

Microadenoma- tumor <1cm

Macroadenoma-tumor >1cm

Rarely can have a large aggressive tumor that invades surrounding structures

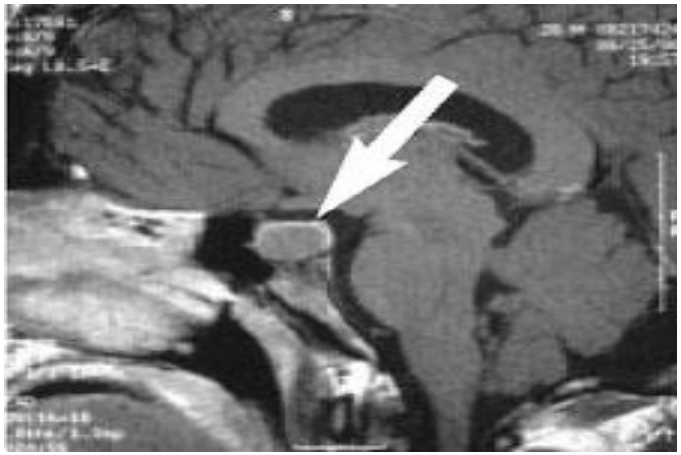
Can be **functional** or **nonfunctional** (producing excess hormone or not)

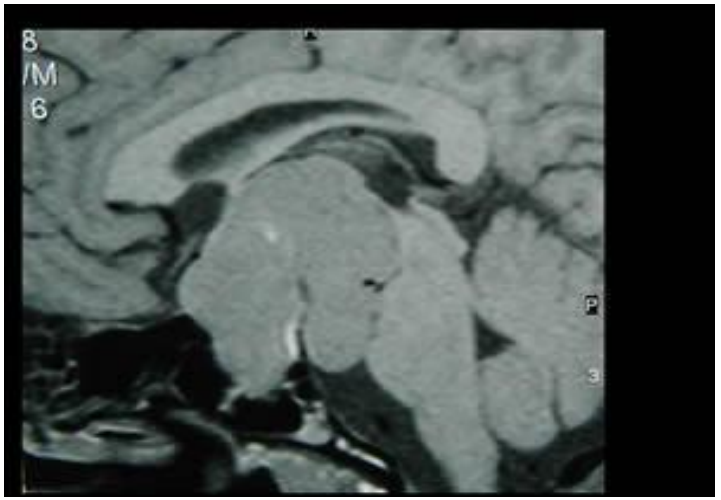
Both small and large tumors, if hormone secreting, can present with signs and symptoms of hormone excess

Pituitary Adenomas
(& definition of terms)

- Large tumors can cause **compression** of surrounding structures, we call these signs and symptoms **MASS EFFECT**
  - Visual loss (bitemporal hemianopsia), cranial nerve palsies
  - Headaches

- Large tumors cause also cause ***loss of function*** of other pituitary hormones - compress or invade the rest of the normal gland
  - **Panhypopituitarism-** is loss of all pituitary hormonal function



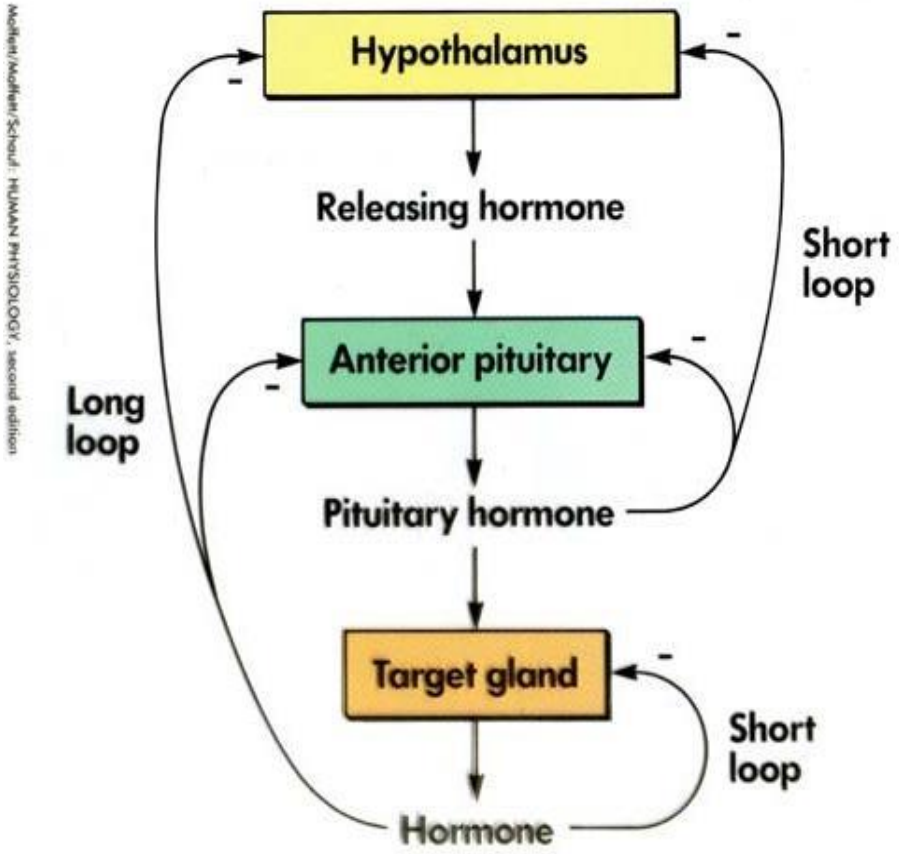


Primary vs. Secondary Hormonal Disorders

Primary- means that the actual target gland is the etiology (ex- primary hypothyroidism-thyroid gland)

Secondary means it is something other than the target gland that is the etiology

- Pituitary dysfunction alters TSH secretion- patient is hypothyroid- but the cause is secondary)



Individual hormones and syndromes of excess

Growth Hormone (GH) Overproduction

Gigantism in children

Acromegaly in adults

RARE

Etiology almost always from pituitary tumor

- Pituitary adenoma- overgrowth of GH secreting cells
  Usually a large tumor (macroadenoma)
- Ectopic GH or Ectopic GHRH secretion- rare (lung, breast, pancreatic or carcinoid tumors)

Growth Hormone (GH) Overproduction

Acromegaly clinical signs and symptoms

- Insidious (mean time from hypersecretion of hormone to actual diagnosis is TEN YEARS!)
- Enlarged hands and feet (ask if ring/shoe size has changed)
- Enlargement of facial bones (increased size of mandible (prognathism) and brow (supraorbital ridges) and widening of the space between teeth
- Sweating (hyperhydrosis)
- Enlargement of the soft tissues (sleep apnea, carpal tunnel)
- Cardiomegaly, hypertension
- Colon polyps/carcinoma? and skin tags
- Diabetes
- Menstrual abnormalities

Untreated acromegalics are at very high risk

for cardiovascular morbidity and mortality







Growth Hormone (GH)

- Released in a pulsatile manner (random values do not give you much information)
- Levels fluctuate throughout life (lab will give you the age specific range)
- In adults, the basal level is very low throughout the day
- Measurement of IGF-1 is the test of choice to screen for overproduction (IGF-1 levels do not fluctuate as much as GH)

Growth Hormone (GH) Overproduction

- Screening lab test- IGF-1
- If elevated, need to confirm biochemical diagnosis…
  - Glucose tolerance test (glucose should suppress GH secretion in healthy people)
    - Give 100 grams of oral glucose, measure GH at 60 minutes and 120 minutes.
    - Levels >1ng/mL- confirms hypersecretion
- Now that have the biochemical diagnosis- can do imaging- pituitary MRI

Acromegaly Treatment

Surgery

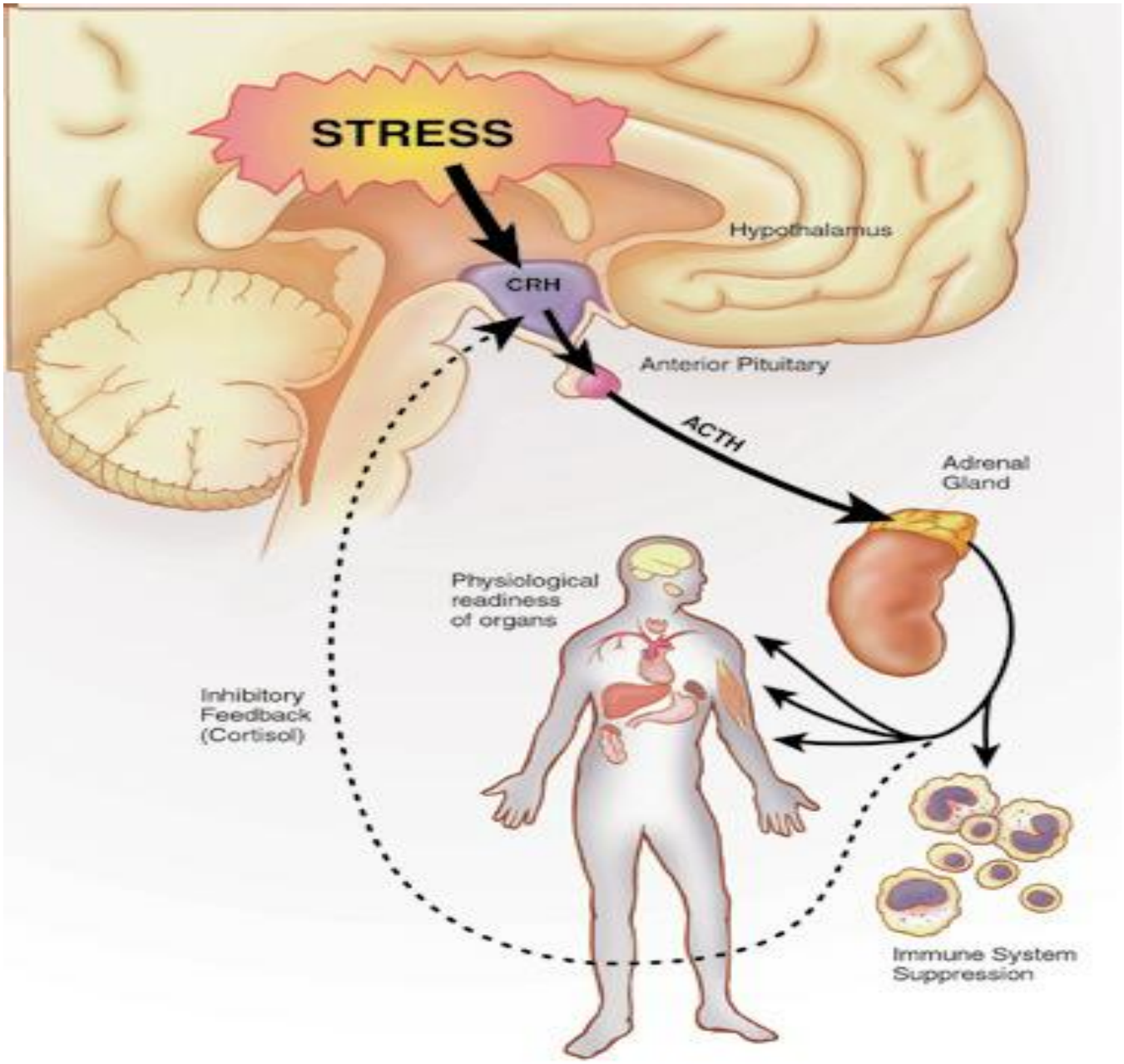
- Transsphenoidal approach if possible

Medical treatment

- Somatostatin (octreotide) inhibits GH
- Dopamine agonists or GH receptor antagonists also used

Radiation

- Used as adjunct treatment to surgery if tumor incompletely removed or recurs


STRESS
Hypothalamus
CRH
Anterior Pituitary
ACTH
Adrenal Gland
Physiological readiness of organs
Inhibitory Feedback (Cortisol)
Immune System Suppression

ACTH overproduction

ACTH overproduction will lead to Glucocorticoid excess.

Glucocorticoid excess leads to leads to a constellation of symptoms and physical features known as **CUSHING'S SYNDROME**

Clinical signs and symptoms:

- Buffalo hump, central obesity, round rosy face, supraclavicular fat pads
- Carbohydrate intolerance, hypertension
- Thin skin, prominent **purple** striae, easy bruising
- Acne, hirsuitism, menstrual disturbances
- Hypokalemia, edema
- Muscle weakness, osteoporosis/osteopenia
- Emotional lability

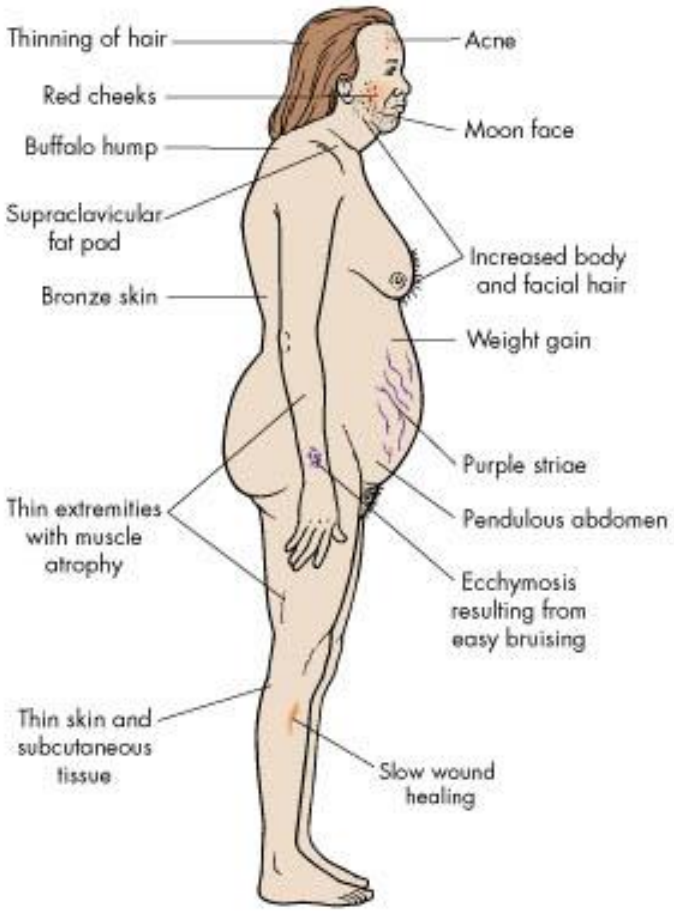




Figure 47-9 Common characteristics of Cushing's syndrome.

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Testing for Cushings

Diagnosis is confirmed with at least two positive screening tests showing cortisol excess:

- **24 hour urinary free cortisol** well above the physiologic range (>150mcg/day)
- **Elevated Late Night Salivary Cortisol**
- **1-mg overnight low dose dexamethasone suppression**:

  1 mg dexamethasone at 11PM, draw cortisol at 8am

  Patients with Cushing's do not suppress their 8AM serum cortisol to below 1.8 ng/dl

  10-20% false-positive results (obesity, depression, alcoholism) => every abnormal test should be confirmed

Cushing's Syndrome

Excess cortisol can be from pituitary, adrenals, or ectopic tissue!

- To distinguish-Check plasma ACTH levels:
  - If < 5 pg/ml => Adrenal Cushing's => CT scan of adrenals
  - If > 20 pg/ml => Pituitary or Ectopic Cushing's
  - If between 5 and 20 pg/ml => Not definitive

It may be difficult to distinguish the different causes of ACTH dependent Cushings

Cushing's Syndrome

ACTH- DEPENDENT (high ACTH levels)

Cushing's disease–means source is the pituitary 80% of endogenous causes

Ectopic ACTH-Non-pituitary tumor makes ACTH

- 5-10% of non-iatrogenic cases of Cushings. ACTH levels usually much higher
- 90+% from chest tumors (carcinoid, small cell lung carcinoma), also medullary thyroid cancer, pancreatic islet cell tumors, pheochromocytoma)
- Can present with signs and symptoms of carcinoma Do imaging to search for site (Chest CT scan)
- First line treatment is surgical resection

ACTH Dependent Cushings

Localize the source of ACTH

High dose dexamethasone suppression test (8mg) or 2 day dexamethasone suppression test

- Measure baseline cortisol, give steroids, then measure cortisol after steroids given
- Expect pituitary source to suppress more

  Rationale is that pituitary ACTH secreting cells may retain more of their negative feedback mechanism

Controversy regarding lab value cut-offs, Many false positives

NOT relied on!!!!

ACTH Dependent Cushings

Localizing source of ACTH

Inferior Petrosal Sinus sampling

Only do this test in a person with confirmed hypercortisolism, when cannot differentiate pituitary vs ectopic ACTH cause

- Pituitary blood drains into the cavernous sinuses, then into the inferior petrosal sinuses
- Sampling blood directly from the petrosal sinuses allows us to calculate the ratio of ACTH coming directly from the pituitary and compare it to peripheral blood
- A ratio >3 is consistent with pituitary Cushing's (Cushing's disease)
- <3 indicates that you should search for ectopic source
- Best done after giving CRH

ACTH overproduction-Cushing's Disease

Cushing's syndrome is called **Cushing's disease** if the cause of glucocorticoid overproduction is from an anterior pituitary tumor overproducing ACTH (the most common cause of Cushings, but still uncommon)

Usually a microadenoma

Diagnosis review-

- Signs and symptoms of cortisol excess
- Confirm cortisol excess on labs tests
- Measure ACTH- will be high in Cushing's disease
- Now do pituitary MRI to search for tumor

Cushings Disease Treatment

- Surgery
  - Transsphenoidal approach if possible
- Radiation
  - Used as adjunct treatment to surgery if tumor incompletely removed or recurs
  - Benefits may not take effect for years
- B/L Adrenalectomy (always irradiate the pituitary before this- concern that pituitary tumor will then grow)
- Medical treatment

  Drugs that block adrenal glucocorticoid synthesis (ketoconazole, metyrapone), somatostatin analogues

TSH

TSH secreting pituitary adenoma is rare

Diagnosis

- Signs and symptoms of hyperthyroidism
- **TSH is normal or slightly high** with thyroid hormone levels also high
  - -**<u>Inappropriate! (expect TSH to be 0 if T4 is high)</u>**
  - -This is Secondary hyperthyroidism

Treatment is surgical resection of pituitary tumor, +/- radiation

- If remove or destroy thyroid gland, you can cause pituitary tumor growth

FSH/LH (Gonadotrophins)

- LH and FSH bind to receptors in ovaries and testes
  - Function- puberty, sex steroid production, reproductive function (ovulation)

- Estrogen, testosterone, inhibin are the hormones made from the gonads in response to LH/FSH
  - Provide negative feedback on gonadotroph secretion
  - Complex concert of events! (release, feedback inhibition, stimulation of pulsatile GnRH, LH, FSH, and sex steroids)

LH/FSH

Gonadotroph secreting tumor

- Rare
- Usually macroadenomas
- Can cause hypogonadism (due to loss of pulsatile secretion)
  - Impotence, infertility, loss of menses
- Rarely can cause increased sex steroid levels
- Treatment is surgical resection +/- radiation

Prolactin

Each hormone we reviewed so far is regulated by hypothalamic hormones (stimulate release), except for **PROLACTIN**- under tonic inhibitory control

- dopamine from the hypothalamus inhibits prolactin) only allowing a low basal secretion in a 'normal' pituitary

The physiologic STIMULUS for prolactin release is pregnancy

- TRH also stimulates release

Function is to stimulate milk production

Prolactin Hypersecretion

Diagnosis

- Signs and symptoms
  - Females-loss of menses (amenorrhea), galactorrhea (lactation), infertility
  - Males- sexual dysfunction (loss of libido, impotence)
  - Prolactin inhibits pulsatility in gonadotroph secretion, leading to estrogen or testosterone deficiency
  - Compressive symptoms and MASS EFFECT if source is pituitary macroadenoma

- Measure serum Prolactin

Prolactin Hypersecretion

Differential diagnosis of high prolactin

- **Pregnancy**
- Renal failure, liver failure
- Primary Hypothyroidism (TRH can stimulate prolactin release)
- Microprolactinoma
- Macroprolactinoma
- Macroadenoma (other large pituitary tumor)
  - Stalk effect
  - Anything disrupting the hypothalamic stalk (mass, infiltration) , the tonic inhibition by dopamine may be disrupted- therefore prolactin elevates
- Hypothalamic disease -loss of tonic inhibition
- Chest wall stimulation, nipple stimulation
- Medications (phenothiazines, metoclopramide)

Remember: pregnancy test, TSH, medication review

Prolactinoma

Most common secretory type of pituitary adenoma

Treatment Indications are: hypogonadism, bothersome galactorrhea, macroprolactinoma, infertility

- Usually first line treatment is medical!
  - Dopamine agonists (cabergoline, bromocriptine)cause tumor shrinkage, can restore menses and fertility
- Surgery can be considered – but only with a highly experienced neurosurgeon and adenoma is well circumscribed (rarely done)

- If fail to respond to medication or **severe** mass effect, then surgery is done

Prolactinoma Review

Different from other pituitary adenomas

- Normally prolactin is tonically inhibited by hypothalamic dopamine
- Large tumors (whether secretory or not) can disrupt this inhibition and elevate prolactin
- First line treatment is usually medical (not surgical)

Non-Secreting (Null cell) Adenoma

- Biologically **inactive** glycoprotein hormones may be secreted
- Usually macroadenomas
- Can present with mass effect and compressive symptoms
- Treatment is surgical +/- radiation

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ADH aka AVP aka Vasopressin and the Term Diabetes Insipidus

- 2022: An international group has recommended that the name "diabetes insipidus" — which in some cases has led to harm — be changed (eliminates confusion with "diabetes mellitus" and reflects pathophysiology)

- **AVP-deficiency (central DI)**
- AVP-resistance (nephrogenic DI)

ADH (AVP) Deficiency (Central DI)

Pituitary or hypothalamic loss of AVP

Have loss of water (Polyuria and Polydipsia, large volumes of **dilute** urine, craves cold water)

Causes: tumor, infiltrative disease, post-surgery, infection, familial, autoimmune, idiopathic

Treat by giving back AVP -causes the urine to concentrate again

- Usually dosed symptomatically by patients
- If patients are able to drink and have access to water, they will be ok, but uncomfortable (constant urination and thirst)

ADH

Excess of ADH

- SIADH (Syndrome of inappropriate secretion of ADH)
- Water retention, can lower serum sodium
- Signs and symptoms- nausea, lethargy, seizures, coma

- **Many causes (nausea, infection, medications, cancer)**
- A hypothalamic tumor making ADH is extremely rare

- Treatment- water restriction, or if severe, hypertonic saline

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Hypopituitarism

- Clinical syndrome resulting from the deficiency of one or multiple pituitary hormones (partial/panhypopituitarism)

- The most common cause is a pituitary macroadenoma (with or without resection)

Pituitary Hormone Underproduction-Hypopituitarism

Causes

- Hypothalamic lesions (disruption of signals to pituitary)
- Pituitary destruction
  - Neoplasm
  - Infection
  - Infiltration (granulomatous disease)
  - Hemorrhage
  - Post-Surgical
  - Congenital or Acquired lesions
  - Radiation

Will cause secondary endocrine gland failure (signals gone!)

Panhypopituitarism

Workup should include:

Endocrine lab testing

MRI of the hypothalamus/pituitary

Treatment depends on the etiology- may include neurosurgical intervention

Hormone replacement

- **ACTH deficient (secondary adrenal insufficiency) a medical emergency- replace glucocorticoid immediately**
- **GH deficiency- GH**
- TSH deficiency- thyroid hormone
- LH/FSH deficient-testosterone/ estrogen
- ADH deficient (AVP-D)-ADH

Pituitary Apoplexy

- Results from hemorrhagic infarction of a pituitary adenoma
- Acute presentation with headaches, change in mental status, ophthalmoplegia and visual loss, sometimes necessitating surgical decompression
- Variable degrees of hypopituitarism

Prolactin Deficiency

Failure of lactation in post-partum women

Sheehan's syndrome- postpartum hemorrhage into the sella

- Usually first sign of Sheehan's syndrome is inability to lactate
- Must test all hormonal axes and provide replacement therapy **(steroids)** as needed

Lymphocytic Hypophysitis

Lymphocytes infiltrate the pituitary / pituitary stalk

- Autoimmune
- May not see mass on MRI (can see stalk thickening)

Usually seen in late pregnancy or postpartum

Associated with hormonal dysfunction

- Loss of anterior pituitary hormonal function
- Diabetes insipidus

Pituitary Incidentaloma

Differential diagnosis: see earlier slide (sellar enlargement). Most commonly it is an adenoma

- Exclude hormonal hypersecretion:
  - PRL, IGF-1, TSH & free T4, screen for Cushing, LH, FSH, Treatment accordingly

- If macroadenoma => also need to rule out hypopituitarism and do formal visual field testing (near optic chiasm). If compression-may go to surgery

- If all testing is normal (nonfunctional, stable) follow with periodic MRI's
  - If macroadenoma, repeat MRI in 6 months
  - If microadenoma, can wait 1 year, then at 2 and 5 years (some don't repeat imaging if <4mm)